

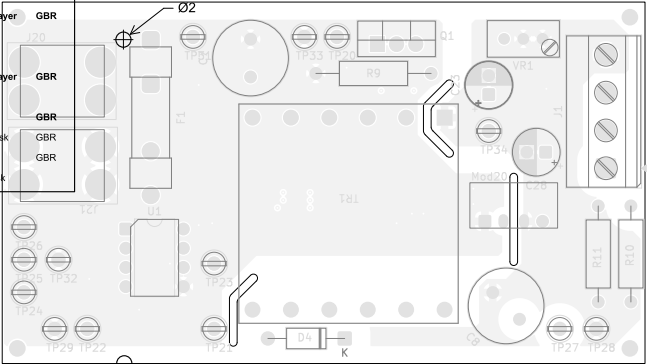
Module Fabrication Document

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Type	Gerber
F.Paste				Paste Mask	
F.Silkscreen				Legend	GBR
F.Mask		0.02mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Ref)	0.07mm (2.00oz)		Plane	GBR
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	Virtual (Top side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
Core		1.21mm	FR4	Dielectric	
Copper	Virtual (Bot side jumpers)	0.035mm (1.00oz)		Copper Layer	GBR
Prepreg		0.1mm	FR4	Dielectric	
Copper	L2 (Sig, PWR)	0.07mm (2.00oz)		Signal	GBR
B.Mask		0.02mm	Solder Resist	Solder Mask	GBR
B.Silkscreen				Legend	GBR
B.Paste				Paste Mask	

Total thickness: 1.66mm
Note: external layer thicknesses are specified after plating

Top Fabrication (Scale 1:1)



Impedance Table

Transmission LIne	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
NA	-	-	-	-	-	-

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING ARE FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR TOP: #8071501A / BOTTOM: GREEN.
- 6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
- 8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
B. Tg 170 C OR EQUIVALENT.
C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY FR.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

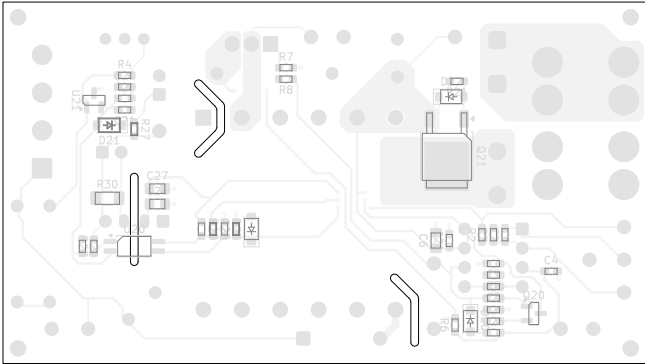
BOARD SIZE85.000 × 47.800 mm
BOARD THICKNESS1.660 mm
TRACE WIDTH0.200 mm
TRACE TO TRACE-0.000 mm
MIN. HOLE (PTH)0.250 mm
MIN. HOLE (NPTH)2.100 mm
ANNULAR RING0.150 mm
COPPER TO HOLE0.254 mm
COPPER TO EDGE0.250 mm
HOLE TO HOLE0.254 mm

All dimensions are in millimeters unless otherwise specified.

	Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR	Variant: PRELIMINARY	Git Hash: 34dba06
		Board Name: Module	Project Name: Isolated step-up 12:450 V	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2024-04-13
	Sheet Path:		Reviewer: NA	Revision: 1.0.0+ (Unreleased)
			Size: A4	Sheet: 1 of 10

Module Fabrication Document

Bottom Fabrication (Scale 1:1)



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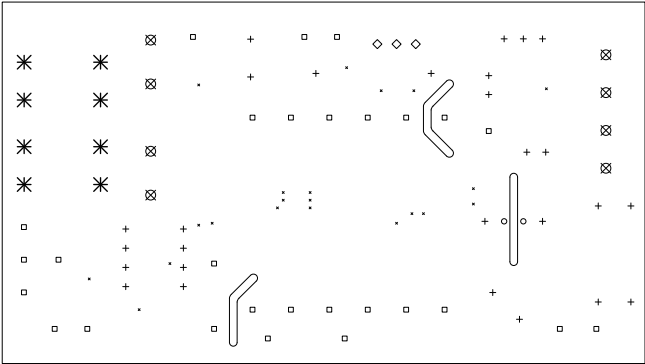
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		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer: NA	Size: A4	Sheet: 2 of 10

Module Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	21	0,25mm (9,84mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Via
○	2	0,70mm (27,56mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
+	27	0,80mm (31,50mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
□	28	0,90mm (35,43mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
◇	3	1,20mm (47,24mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⊠	8	1,30mm (51,18mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
⋈	8	1,80mm (70,87mils)	PTH	Round	L1 (Ref) - L2 (Slg, PWR)	Pad
Total 97						

Drill Drawing L1 - L4 (Scale 1:1)



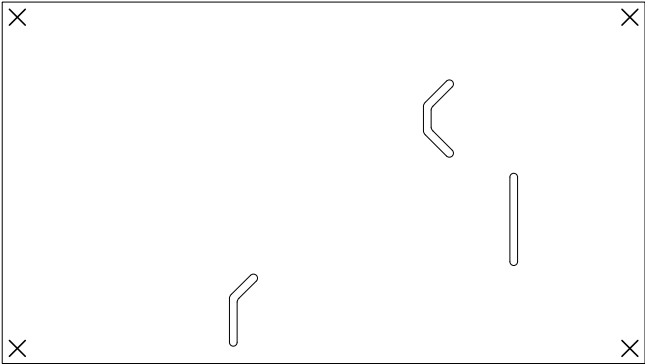
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	Comment 3				
	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Drill Drawing (L1 - L4)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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			NA	A4	3 of 10

Module Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	2.10mm (82.68mils)	NPTH	Round	L1 (Ref) - L2 (Sig, PWR)	Mechanical
	Total 4					

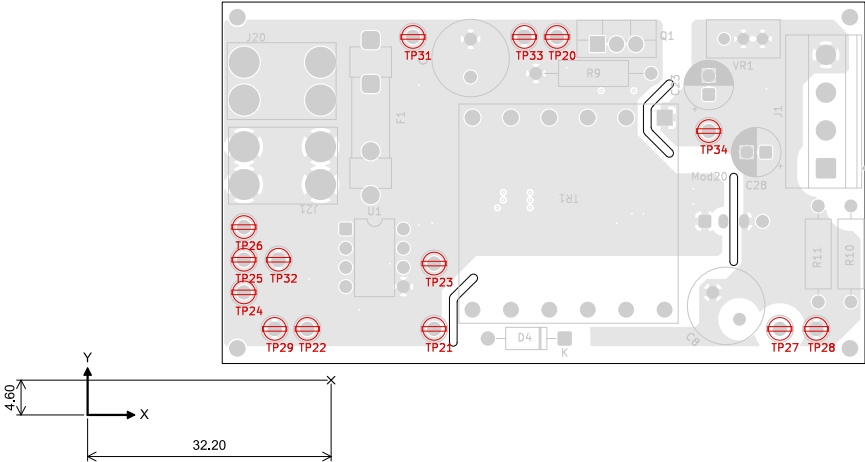
Drill Drawing L1 - L4 (Scale 1:1)



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	Comment 3	Module		Isolated step-up 12:450 V	
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	Drill Drawing (L1 - L4)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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			NA	A4	4 of 10

Module Fabrication Document

Top Test Points (Scale 1:1)



Ref.	Net	X [mm]	Y [mm]
TP20	SW	62.07	49.99
TP21	Src	45.82	11.38
TP22	Isns	29.05	11.38
TP23	PWM	45.82	20.02
TP24	Vfb	20.67	16.21
TP25	Comp	20.67	20.53
TP26	OpEm	20.67	24.84
TP27	U500_sec	91.54	11.38
TP28	Sec_GND	96.36	11.38
TP29	GND	24.73	11.38
TP31	Vin_p	43.02	49.99
TP32	GND	25.24	20.53
TP33	GND	57.75	49.99
TP34	U5_sec	82.14	37.54

Ref.	Net	X [mm]	Y [mm]
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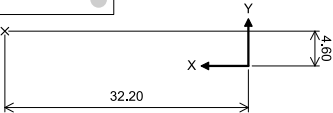
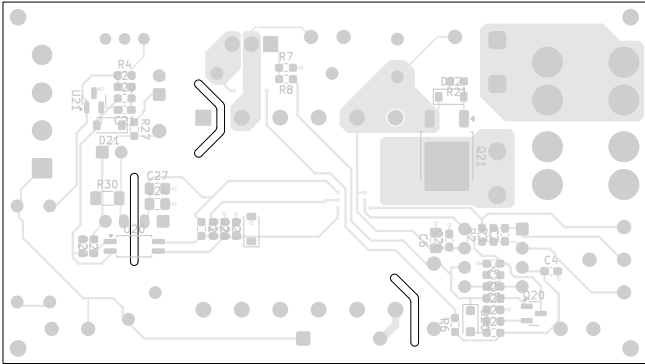
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	Comment 3				
	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Top Test Points (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
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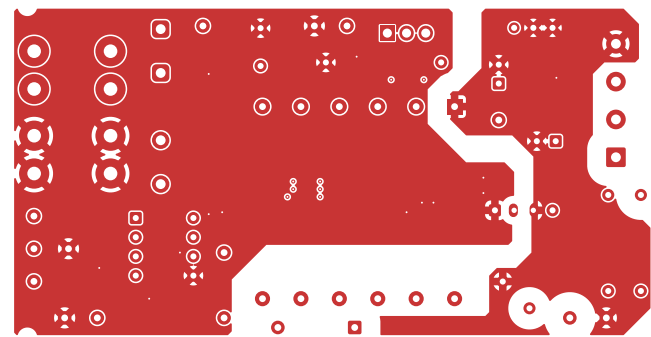


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	Comment 4	Module			
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	Bottom Test Points (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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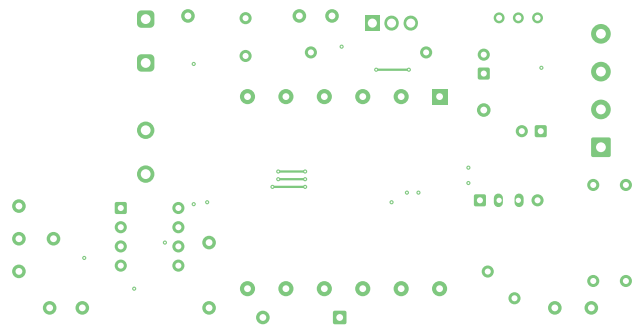
L1 (Ref) (Scale 1:1)



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	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
	Comment 4				
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	L1 (Ref) (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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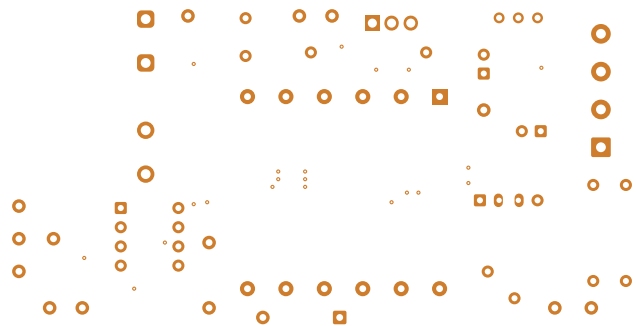
Virtual (Top side jumpers) (Scale 1:1)



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	Comment 2	Board Name:		Project Name:	
	Comment 3	Module		Isolated step-up 12:450 V	
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	Virtual (Top side jumpers) (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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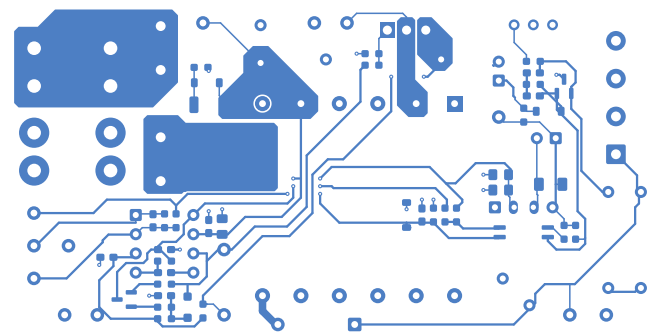
Virtual (Bot side jumpers) (Scale 1:1)



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	Comment 3				
	Comment 4	Module		Isolated step-up 12:450 V	
	Sheet Title:	File Name:	Designer:	Date:	Revision:
	Virtual (Bot side jumpers) (Scale 1:1)	Isolated_12to450V.kicad_pcb	FR	2024-04-13	1.0.0+ (Unreleased)
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Module Fabrication Document

L2 (Sig, PWR) (Scale 1:1)



	Comments: Comment 1 Comment 2 Comment 3 Comment 4	Company: FR		Variant: PRELIMINARY	Git Hash: 34dba06
		Board Name: Module		Project Name: Isolated step-up 12:450 V	
	Sheet Title: L2 (Sig, PWR) (Scale 1:1)	File Name: Isolated_12to450V.kicad_pcb	Designer: FR	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
	Sheet Path:		Reviewer: NA	Size: A4	Sheet: 10 of 10